MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND, PRECISION,

STYLE RR4000

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for style RR4000, variable, wire-wound, precision resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

Single-section (cup) resistors

RR4000	C	2	A	7	J	101
Style (1.2.1.1)	Function and shaft length (1.2.1.2)	Class (1.2.1.3)	Resistance temperature characteristic (1.2.1.4)	Rotational life characteristic (1.2.1.5)	Function conformity tolerance and resistance tolerance characteristic (1.2.1.6)	Resistance (1.2.1.7)

1.2.1.1 Style RR4000. Style RR4000 is applicable to this specification.

1.2.1.2 Function and shaft length. The function and shaft length applicable to this specification shall be identified by a single symbol in accordance with table I.

TABLE I. Function and shaft length.

Symbol	Shaft length Tolerance ±1/32 (.79)	Function
C - - -	5/8 (15.88)	Linear
D - - -	3/4 (19.05)	Linear
E - - -	7/8 (22.23)	Linear
F - - -	1 (25.40)	Linear

1.2.1.3 Class. The class applicable to this specification shall be symbols 2, 3, 4, and 5.

1.2.1.4 Resistance temperature characteristic. The resistance temperature characteristics applicable to this specification are symbols A to H, inclusive.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Electronics Research and Development Command, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.5 Rotational life characteristics. The rotational life characteristics applicable to this specification are symbols 7, 8, and 9. This potentiometer is capable of operation up to 100 rpm.

1.2.1.6 Function conformity tolerance and resistance tolerance characteristics. The function conformity tolerance and resistance tolerance characteristics applicable to this specification are symbols G to L, inclusive and S to X, inclusive.

1.2.1.7 Resistance. The nominal total resistance values applicable to this specification shall be in accordance with table II.

TABLE II. Nominal total resistance and nominal resolution characteristics.

Nominal total resistance value	Nominal resolution
(Ohms)	(Percent)
100	.07
200	.06
500	.05
1,000	.045
2,000	.035
5,000	.025
10,000	.019
20,000	.016
50,000	.012

2. APPLICABLE DOCUMENTS

* 2.1 Government specification. Unless otherwise specified, the following specification of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-12934 - Resistors, Variable, Wire-wound, Precision, General Specification for.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-12934 and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Minimum resistance. Minimum total resistance value shall be 100 ohms.

3.4 Maximum resistance. Maximum total resistance value shall be 50,000 ohms.

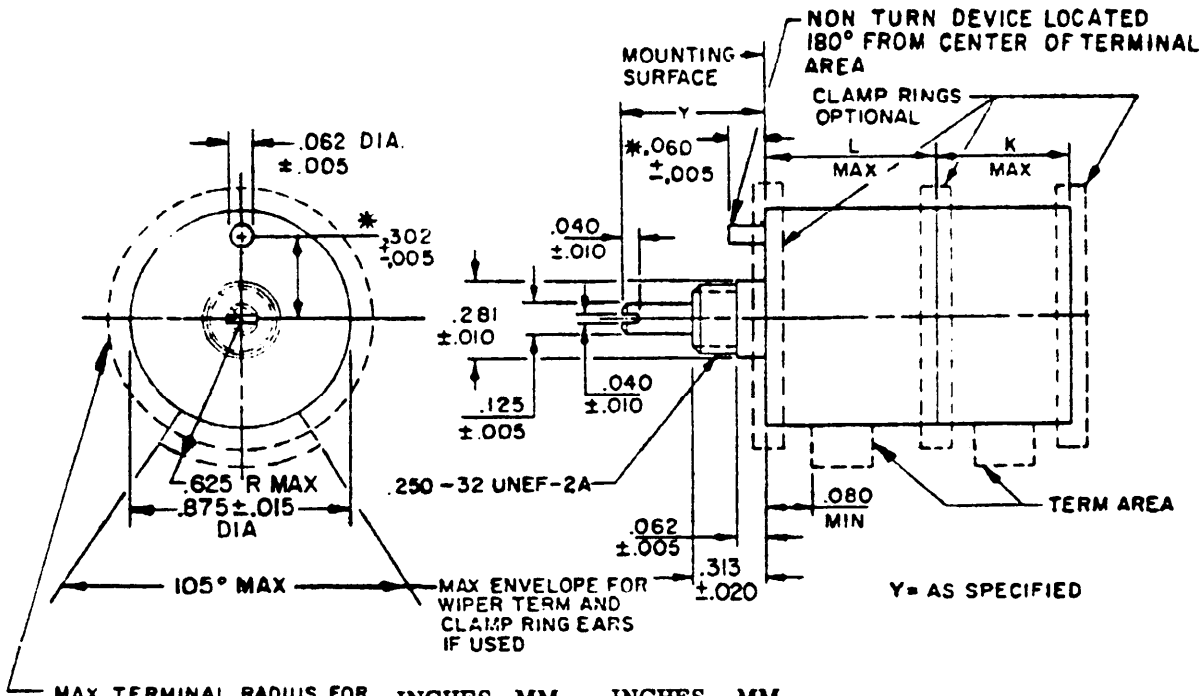
3.5 Mechanical travel. Mechanical travel shall be 3,600 $^{+10}_{-0}$ degrees.

3.6 Actual electrical travel. The actual electrical travel shall be 3,600 $^{+4}_{-0}$ degrees.

3.7 Ganged cups. There shall be no more than 2 cups ganged.

3.8 Phasing. For independent linearity, the phasing between the cups shall be within ±1 degree. This shall be measured, with respect to the first cup, starting from the counterclockwise end point.

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INCHES	MM	INCHES	MM
.005	.13	.080	2.03
.010	.25	.125	3.18
.015	.38	.250	6.35
.020	.51	.281	7.14
.040	1.02	.313	7.95
.060	1.52	.625	15.88
.062	1.57	.875	22.23

Style	Turns	L (max)	K (max)	Maximum continuous working voltage (volts)	Power rating (watts) 1/	Maximum starting and running torque (ounce-inch)				Stop-torque (pound-inch)
						Ten-turn, single cup		Per each additional ten-turn cup		
						starting	running	starting	running	
RR4000	Ten	1.500 (38.10)	1.500 (38.10)	300	2.0	.7	.6	.6	.5	3

1/ When ten turn units are ganged all cups will be 75% of rated wattage.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 1. Style RR4000.

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3.9 Function. The type of function applicable to this specification is linear function (independent linearity).

3.10 Weight. The maximum weight shall be .09 pound and for each additional cup .06 pound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-12934 and as specified herein.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-R-12934.

6. NOTES

6.1 Notes. The notes specified in MIL-R-12934 are applicable to this specification.

6.2 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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RESISTORS, VARIABLE, WIRE-WOUND, PRECISION, STYLE RR4000

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